

Choice as Public Policy: Evidence from Optional Workers' Compensation Coverage in Texas

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Abstract

Using unique administrative data from Texas, I study employers' decisions to opt out of the workers' compensation system. I find that employers in unsafe industries do opt out and, in fact, less safe employers within some industries are more likely to opt out. If employers decide to opt in, they limit liability from employee injury suits but pay a premium. I examine how employers respond to changes in two premium components: bureau-set loss costs and insurer-specific loadings. While employers are generally responsive to both, those with low-injury occupations are much more price-sensitive. This is because insurers use loadings to offset perceived overcharges from loss costs, mainly in low-injury occupations. For firms with high-injury occupations, what then drives the choice to opt out, if not premiums? I find opting out allows them to avoid fully raising wages to match the expected injury costs shifted to employees.

Keywords: workers' compensation, public policy, insurance, coverage mandates

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1 Introduction

Across the US, nearly every state mandates that employers provide their employees workers’ compensation coverage, which pays benefits for work-related injuries and illnesses. Texas is the one state in the US with a robust employer opt-out. When employers opt in to the workers’ compensation system, they agree to pay claims on a uniform basis but also limit the ability of employees to sue them in court. By contrast, when employers opt out, they open themselves up to tort suits from injured workers. Thus, this coverage decision, made by employers, also directly affects workers. Despite the fact that legislators in several states¹ have considered efforts to remove workers’ compensation coverage mandates, little evidence exists as to what components of this system may push employers in or out; this paper’s results help to fill this gap in our understanding.

Despite a lack of attention in the recent economics literature, workers’ compensation systems are quite notable, paying more in benefits in 2022 (roughly \$62 billion) than unemployment insurance programs and about the same amount as the earned income tax credit (Welch et al., 2024). In Texas, which is the focus of this paper, nearly \$1.4 billion in benefits were paid by employers opting in to the system in 2022. Texas is the focus of this paper given it has the only robust opt out option for employers; outside of Texas, it is not possible to study the effect of not being covered by workers’ compensation, both because of data access and the fact that only small employers (between one and five employees) may be able to take a coverage exemption. Nationally, about 92% of all employment was covered in 2022 (Welch et al., 2024). In Texas, however, 25% of employers opt out of the system, employing 17% of employees; this amounts to more than 1.5 million employees working for firms that opt out (DWC, 2022).

In every US state, an employer is considered “covered” when they provide workers’ compensation coverage to employees, either via buying insurance or organizing a self-insurance

¹Oklahoma, Tennessee, Florida, Arkansas, and South Carolina have all proposed bills to do so, with Oklahoma even passing one before the state Supreme Court found it unconstitutional. The Department of Labor (DOL) has recently expressed concern with this trend: <https://tinyurl.com/yhcnn97z>.

arrangement certified by state regulators (where it is allowed). By being covered, the employer and its employees are “subscribers” to the workers’ compensation system defined by two key pillars: 1) employees are given nearly guaranteed payment for injuries at work in the form of lost income replacement (not full) and medical care, and 2) employers are generally protected from employees filing lawsuits for larger damage amounts (i.e., workers’ compensation is the exclusive remedy available to employees). This is the purported “grand bargain” of workers’ compensation systems. Thus, in Texas, the choice to be a subscriber lives with the employer, and when an employer is a non-subscriber, they lose their legal protections and can be sued by their employees (even if they offer alternative injury plans, generally).²

In this setting, employers opt out of coverage if the cost of opting in exceeds the cost of opting out. That is, the cost of workers’ compensation coverage (or to setup self-insurance) exceeds the sum of expected costs from: employee tort suits, the provision of informal work injury plans, and increased employee wages in response to risk shifting. The ultimate guiding question of this paper is: which of these factors push employers on the margin of opting in or out of the workers’ compensation system? This question is unanswerable completely, given the lack of information on tort (and settlement) costs outside of the system. Nonetheless, I aim to provide insight into this broader question by asking several more specific research questions focused on the Texas setting. I first consider the cost of opting in, and I ask: which elements of the insurance premium matter to employers (given each is regulated differently)? I then focus on the measurable element of opt out costs and ask: do firms avoid complete pass-through of work injury risk to higher wages for workers? In comparing these costs, it is ambiguous whether employers in higher-injury industries will opt out and, if they do, whether their factors differ. Thus, I also ask: 1) do employers in unsafe industries opt out and, if so, 2) do the premium and pass-through factors vary by industry or occupation safety?

To answer these research questions, I make use of a unique administrative dataset con-

²Throughout the paper, I use “subscriber” and “covered” interchangeably, referring to employers who purchase workers’ compensation insurance or arrange self-insurance. Employers who provide alternative injury plans are deemed “uncovered” or called a “non-subscriber” because they do not “subscribe” to the exclusive remedy rules.

structed via several open records requests in Texas (similar in nature to [Cabral et al. \(2022\)](#)). In addition to providing occupation-level (within an employer) information on payroll and insurance costs among all employers purchasing workers' compensation insurance over the 2015-2021 period, the main data also include detailed information on each worker injury claim. I supplement these data with summary information on injury counts, across various dimensions, among opt-out employers. Finally, I collect information on the universe of self-insured firms to complete a full picture of subscribers to the system in Texas.

First, I document that statewide covered rates (i.e., the percentage of payroll paid by subscribers) in my administrative data track those reported by the government's employer surveys. After doing so, I show that there is significant variation across industries and over time in covered rates. The descriptive evidence I present suggests substantial levels of opt-outs in unsafe industries and, within some industries, injury rates are higher among firms opting out than opting in. One of the three prior papers in this space, to my knowledge, documents that there is limited levels of adverse selection, using variation in a bureau-set component of premiums within an occupation group, to inspire a coverage mandate ([Cabral et al., 2022](#)). My descriptive results, combined with the findings of [Cabral et al. \(2022\)](#), motivate an understanding of how price is decomposed and how the component parts may shift firms employing different types of workers around distinctly.

In line with that motivation, I then separate out the first step of the workers' compensation premium algorithm, given it applies to all firms. Specifically, I separately study bureau-set loss costs, which are intended to reflect expected loss and loss adjustment expenses for a given occupation in Texas, and insurer-specific loadings charges, which allow insurers to modify the bureau rate and add in charges for profit margins, residual loss uncertainty, and operating expenses. I show that the two components are negatively related and then largely replicate the baseline finding of [Cabral et al. \(2022\)](#) with respect to demand estimates derived from changes in loss costs. When I add in loadings, firms appear more sensitive to loadings than to loss costs. The results also indicate that price sensitivity is

very different among firms with employees in high- and low-injury occupations. While there is almost no price sensitivity among employers with employees in high-injury occupations, their low-injury counterparts are quite price sensitive, with variation by element. Namely, a 1% decrease in loadings generates a 0.84% increase in covered payroll while a 1% decrease in loss costs generates only a 0.42% increase in covered payroll. Ultimately, I provide evidence that this large sensitivity to loadings comes from insurers' downward adjustments of loss costs in occupations that are perceived to be overcharged by the bureau.

The other two prior papers in this space focus on opt out firm injuries. [Butler \(1996\)](#) shows that injury rates are higher among opt-out firms than opt-in firms in some industries. [Jinks et al. \(2020\)](#) show that employers opting out save 50% on injury costs, mostly through informal injury plans that reject more claims and provide less generous benefits. With these papers as motivation, my findings on opting out in unsafe industries and limited price sensitivity among firms with high-injury occupations suggest that there may be other, salience-related motivations to opt out. Thus, I focus on one of those avenues: wage pass-through. I estimate that observed wage increases in response to losing coverage suggest much less than complete (and often less than 50% complete) pass-through. This is true after accounting for the percentage of opt-out firms that provide alternative benefit plans and their associated generosity (using the estimates from [Jinks et al. \(2020\)](#)). I then show that pass-through in high-injury industries is far less than it is in low-injury industries; in fact, in low-injury industries, pass-through is nearly complete, while firms in high-injury industries achieve average pass-through that is less than 50% of what would be considered complete.

This project contributes to several strands of literature. First, this paper contributes to the literature focused on the operation of workers' compensation markets with optional coverage. As discussed above, the only state with a robust opt-out is Texas, with the last state to make the change to mandatory coverage being South Carolina in 1997 ([Morantz, 2010](#)) and all other states mandating coverage at least since 1987 ([Nelson, 1990](#)). I add to this space alongside the three papers used to motivate above ([Butler, 1996](#); [Jinks et al.](#),

2020; Cabral et al., 2022) by focusing on the active opt out margin, rather than realized experiences of opt out firms (Butler, 1996; Jinks et al., 2020), and by decomposing elements of price (Cabral et al., 2022).

In decomposing elements of the workers’ compensation premium, this paper also contributes to the literature on pricing in workers’ compensation insurance. Given pricing algorithms are complex, this literature is quite broad. Roberts (2005) provides an extensive overview of this space. Yet, Cabral et al. (2022) were the first authors to empirically estimate the demand for workers’ compensation insurance coverage. Given the authors use loss costs, I introduce loadings to demand in this literature for the first time empirically.³ The price results I present are also policy-relevant in identifying heterogeneous responses to different regulatory levers within prices, as prior work has shown the (sometimes catastrophic) result of regulating workers’ compensation insurance premiums (Harrington and Danzon, 2000; Barkume and Ruser, 2001; Danzon and Harrington, 2001).

This paper also contributes to the body of work focused on the relationship between workers’ compensation programs and wages. Several papers provide important results about how workers’ compensation benefits shift wages that are consistent with compensating differentials⁴ (Viscusi and Moore, 1987; Moore and Viscusi, 1990a,b; Kaestner and Carroll, 1997; Gruber and Krueger, 1991; Fishback and Kantor, 1995).⁵ Gruber and Krueger (1991) present the most renowned work in this area, showing that marginal benefit and wage changes offset nearly dollar for dollar.⁶ All of this work focuses on the wage effects of either changing benefits on the margin or creating programs for the first time; my work adds to this space

³I note that Schmidle (1995) discussed how insurer adjustments to loss costs allows for the optimal contract.

⁴The theory of compensating differentials between non-wage job characteristics and wages has been explored widely, with mixed evidence and some (surprising) relationships (Brown, 1980; Sorkin, 2018; Wissmann, 2022). There is some notable ambiguity in settings where labor may not function properly, such as employee miscalculation of risk or lack of workers’ compensation salience (Smith, 1979). Moss (2002) concurs in arguing for government involvement in areas where individual misunderstanding of risk is present. Compensating differentials relating to quit rates in the work context have also been explored (Viscusi, 1979), as have changes in employers’ wage costs (Holmlund, 1983).

⁵For interested readers, Krueger and Meyer (2002) review empirical evidence in workers’ compensation, including the effects of benefit level changes on injuries, on both the intensive and extensive margins.

⁶Their analysis forms the crux of the argument that Cabral et al. (2022) use to justify their welfare approach.

by focusing on the effects of removing coverage and framing it as a reason for employers to consider opting out.

Finally, this work contributes to the broader literature on the use of insurance coverage mandates in public policy, which is directly related to how we think choice plays a role in optimal insurance settings, an application of choice in policy-making more broadly (Chetty, 2015; Bhargava and Loewenstein, 2015; Viscusi and Gayer, 2015). Coverage mandates, which are inspired by the (lack of) role of choice, have been widely studied in health insurance (i.e., Gruber and Madrian (1994); Auerbach et al. (2010); Kolstad and Kowalski (2016); Barkowski et al. (2020); Richards and Tello-Trillo (2021); Barkowski and McLaughlin (2022); Maclean and Webber (2022)), but little empirical evidence exists in other insurance settings (Wriggins (2012) discusses auto insurance). Thus, my analysis builds on Cabral et al. (2022) to ask about the consequences of choice for employers.

Given the history of workers’ compensation and the nuance of the Texas system, Section 2 describes the institutional background. Section 3 describes the data, while Section 4 provides the framework through which I think about firms opting in and opting out. Then, Section 5 presents several descriptive facts. In understanding more about the specific motivations of opting in or out, Sections 6 and 7 deal with the cost of opting in (prices) and one cost of opting out (wage pass-through), respectively. Finally, Section 8 presents discussion of the results and concludes.

2 Institutional Background

Workers’ compensation was the first social insurance program in the US. As a state-based system, the first workers’ compensation programs were created starting in 1911, long before federal social insurance programs were created.⁷ Each state has its own workers’

⁷It is important to define social insurance in this context, given several definitions have been proposed. The National Academy of Social Insurance defines social insurance as follows: “Social insurance encompasses broad-based systems that help workers and their families pool risks to avoid loss of income due to retirement, death, disability, or unemployment, and to ensure access to health care” (via the Academy’s website: <https://www.nasi.org/about/>). This definition may be problematic given it doesn’t rule out transfers of resources across individuals. Chetty and Finkelstein (2013) recognize that and define social insurance as “...government programs that transfer resources across states of nature after an individual is born rather than transfers of resources across individuals (e.g., through re-distributive taxation).” This definition is

compensation insurance market, sets its own benefits levels by statute, and makes its own decisions regarding whether to make coverage compulsory. In each state except for Texas, workers' compensation coverage is required for employers to provide to their employees, which has been true for the last several decades.⁸ In this setting, the employer organizes the coverage arrangement and every employee within a state receives the same protection. Workers' compensation benefits provide payment for lost income (generally replacing two-thirds of one's wage) and medical benefits (including rehabilitation costs) due to a work-related injury or illness.

2.1 Historical Development

Before workers' compensation programs were created, a lawsuit against one's employer was the only way to be compensated for one's work injury costs, and employees were quite likely to lose these lawsuits. This was generally due to the fact that employers could use common law defenses to win against most employee suits. These defenses include: 1) claiming the employee may have caused the accident on his/her own, 2) claiming a fellow employee was at fault, and/or 3) claiming that the injured employee had "assumed the risk" of injury by accepting the offer of employment. In the early 1900s, even before workers' compensation programs were created, states generally moved to eliminate these defenses for employers.

When workers' compensation programs were eventually created, they were largely considered the "grand bargain." This bargain refers to items given up by both employers and employees to design the workers' compensation system with two key pillars: 1) employees are

adequate but would not exclude payments from programs like the National Flood Insurance Program or Terrorism Risk and Insurance Program, which are typically not considered as social insurance programs. [Krueger and Meyer \(2002\)](#) offer a more refined version and define it as "...compulsory, contributory government programs that provide benefits to individuals if certain conditions are met." This definition would exclude transfer programs like Medicaid and Supplemental Security Income, while at the same time suggesting that optional workers' compensation programs are not social insurance. This nicely highlights the importance of this study in understanding more about deviating from social insurance principles. The definition I proffer and use as a guide combines the best elements of these to suggest that *social insurance programs are programs which, with government intervention, provide statutorily defined benefits to workers via a compulsory program that is funded by contributions of the employer, employee, or both and exists to transfer resources across states of nature and not across individuals.*

⁸Wyoming and South Dakota are known to allow for limited opt-outs compared to Texas but more so than other states, where it is common only for very small firms to opt out.

given near guaranteed payment for injuries at work in the form of lost income replacement and medical care; 2) employers are generally protected from employees filing lawsuits for larger damage amounts. On the first element, as long as employee can show that a work injury was an accident arising out, and in the course, of employment, then they will be paid. This makes workers' compensation a strict liability system. Employers, on the other hand, gave up the ability to robustly reject claims, even if the employee made the accident substantially more likely. On the second element, an employee is not able to choose between workers' compensation benefits and suing his or her employer for negligence. This makes workers' compensation the only remedy available for work injuries, which forces the employee to give up their chance to win more in court. Insurers also joined in on this bargain, happy to use coverage mandates as a way to expand their coverage options (Fishback and Kantor, 1998).

Not all scholars agree, however, that all three parties were eager to make this bargain. Weinstein (1967), for example, suggests that many employers became eager to force the workers' compensation system on employees after states removed the common law defenses described earlier, primarily to stem a burgeoning of employee tort success. This dynamic has been recognized by others (Burton and Berkowitz, 1971) and has been the impetus for deep questions about whether workers are really equal winners with this bargain in the present day (Amchan, 1983; Spieler, 2017).

2.2 The Texas Setting

In Texas, employers have the unique choice about whether to be covered or uncovered with respect to this system, and the employee chooses only where to work (assuming to be maximizing over a decision set that includes this information).⁹ Employees do not have access to workers' compensation insurance as a product to buy themselves. Coverage, in

⁹Interestingly, section 406.034 of the Texas Labor Code gives an employee the option to opt out of workers' compensation coverage at their employer and retain the right to pursue negligence claims. Employees must elect this, in writing, within 5 days of being covered (either from the hire date or the date of coverage if the firm is newly insured). This only exists as an option for employees at insured firms. Data as to this option remains unavailable via open records request.

Texas and every state, can be obtained through the purchase of insurance or the arrangement of a certified self-insurance scheme (to be approved by the state for financial soundness).¹⁰ Either of these arrangements is required to pay benefits to injured workers as prescribed by the statutory benefit levels in a given state, which may change over time. When an employer opts to be uncovered in Texas, they are also known to be a “non-subscriber,” which refers to the fact that uncovered employers are not subscribed to the “grand bargain” discussed above, meaning the employee may file a tort suit against the employer for more than what subscribers are mandated to provide per statute. The employee could also lose that suit and receive less (including nothing) than statutory levels.

A caveat, however, does exist in that uncovered employers may still provide alternative injury plans (broadly governed by the Employee Retirement Income Security Act of 1974, or ERISA for short) to their employees for on-the-job injuries. The Division of Workers’ Compensation (DWC) at the Texas Department of Insurance (TDI) releases reports about non-subscription, with evidence that of the 17% of employees that worked for non-subscribed, private-sector employers in 2022 (nearly 1.5 million employees), 73% were provided an alternative injury plan (DWC, 2022). This implies that roughly 5% of all Texas private-sector employees (nearly 500,000 workers) operate in an environment where the *only* compensation mechanism for work injuries is through a lawsuit. The percentage of employees working for non-subscribers without alternative injury plans has increased over time, from 12% in 2004 to 18% by 2010 to the 2022 value of 27%.

When they are offered, these plans tend to be more strict, which is in keeping with the evidence from Jinks et al. (2020) that non-subscription is a cost-cutting endeavor. Strictness may mean several things. First, covered services are more limited. While most workers’ compensation systems in Texas cover acupuncture and chiropractor services (claim instructions are even readily available online for the largest insurer: Texas Mutual), only 20% and 13% of alternative plans cover these, respectively (DWC, 2022). Second, reporting requirements

¹⁰Ohio and Wyoming are the only two states that do not allow for self-insurance; these states also operate monopolized insurers that provide virtually all of the workers’ compensation insurance to their employers.

are much more strict. Among subscribers in Texas, employees generally have 30 days to report an injury; in 2022, 57% of non-subscribers require same day reporting with another 32% requiring reporting within 24 hours or by the next day. Third, and finally, benefits are more limited. For instance, among the nearly 30% of non-subscribers who offer alternative plans, only 67% pay medical benefits for as long as is necessary and only 47% pay lost income replacement benefits. Death benefits are always paid to a widow(er) and dependent children among Texas subscribers; yet, among non-subscribers, 71% offer death benefits to the spouse and just 27% to dependent kids.

Thus, in this non-subscriber world, employees may be forced to manage substantially more risk, both by taking more limited benefits with less-uniform rules (that cannot be easily searched as statewide rules, for example) and through navigating potential tort claims (and hiring necessary legal counsel). Employers, too, could face more risk if the lawsuit avenue is 1) salient and 2) used efficiently, given their common law defenses are still restricted.

One concern in this setting is the use of binding arbitration agreements, which have become more prominent and increasingly upheld by courts ([Sargent et al., 2020](#)). In 2020, 15% of non-subscribers used these to force an employee into arbitration rather than allowing them to pursue a negligence claim in court ([DWC, 2020](#)). 32% of non-subscribers do not pay benefits at all to employees who do not agree to binding arbitration, while 37% appear to make it a condition of employment; 35% of non-subscribers require employees pay all or some of the eventual arbitration costs. These descriptive facts highlight additional potential concerns within the system for workers, but they also highlight that negligence suits remain open to many of the employees who are employed by non-subscribers, which is relevant to the calculus of costs, benefits, and salience.

3 Data

This project combines data from several sources. The unique administrative data comes from several open records requests in Texas, four of which were fulfilled by TDI and one by

DWC.¹¹ I have also gathered publicly-available data from the Bureau of Labor Statistics’ (BLS) Quarterly Census of Employment and Wages (QCEW) and Survey of Occupational Injuries and Illnesses (SOII).

3.1 Open Records Data

There are three relevant components to the open records data. The first, and most comprehensive, dataset focuses on the universe of subscribers in the Texas system between 2015 and 2021.¹² These data are linked across subscribed employers over time but remain randomized to protect employee identities. For each firm (employer), the data include information on premiums paid, zip-code, and overall injury costs and counts. The main employer-level data is then linked to more detailed occupation-level and injury-level information.

In the occupation-level data, there is information on payroll and rates. Importantly, one can establish - across employers - the total amount of payroll covered within an occupation and year. With rate information at the employer-occupation-year level, one can also decompose the elements of price (which is done in the next section). In the injury-level data, there is information on every workers’ compensation claim among subscribers, derived from roughly 1.1 million unique claims. The data provide information on benefits paid (broken down by medical and lost income), projected benefits to be paid in the future, the nature of injury, indicators for settlements, and amounts paid for legal representation.

The second component of the open records data is summary-level information on injuries at non-subscribed employers. These data, spanning 2010-2022, represent injury counts along various dimensions, including race, gender, ethnicity, industry, and type of injury. While all non-subscribers are required to report each work injury or illness, there may still be under-reporting. Nonetheless, these data suggest several descriptive facts, which are reported in panel (a) of Table 1 (only 2015-2021 data is reported, which aligns with the baseline period

¹¹The author is willing to share resources on duplicating such open records requests, as was done by [Cabral et al. \(2022\)](#) through their publication materials.

¹²A detailed presentation of summary statistics for these administrative data is saved for the next section, where I motivate the forthcoming analysis using these data. I do not use the other open records data in that motivation so present some baseline statistics for those pieces below (in this section).

from the subscriber data).¹³ Most reported non-subscriber injuries are among females, which is consistent across the period. The portion of injuries pertaining to White and Black workers has increased over time, by 1.5 percentage points each. At the same time, the portion of injuries coming from Hispanic workers has decreased by roughly 3.5 percentage points. The prevalence of disease cases has risen over time, largely due to a persistent number of disease cases while overall reported injury numbers fell. The same is true for fatalities.

Table 1: Non-Subscriber and Self-Insured Data

	(1) 2015	(2) 2016	(3) 2017	(4) 2018	(5) 2019	(6) 2020	(7) 2021
Panel (a): Non-Subscriber Injuries and Illnesses							
Female	56.64%	56.37%	57.46%	57.90%	57.86%	62.75%	58.98%
White	48.69%	49.21%	48.70%	49.39%	47.28%	50.26%	50.33%
Hispanic	36.45%	34.77%	34.56%	35.05%	35.38%	33.48%	32.93%
Black	12.43%	13.26%	13.16%	12.96%	14.70%	13.52%	14.09%
Disease-related	7.59%	9.19%	9.30%	10.14%	7.01%	24.08%	12.48%
Fatality	0.10%	0.14%	0.11%	0.11%	0.17%	0.09%	0.19%
Count	10,407	9,747	10,914	11,410	10,451	11,627	9,406
Panel (b): Self-Insured Employers							
Firms	41	41	38	35	35	33	31
Establishments	266	260	230	207	217	219	179
Employees	145,154	145,139	133,238	130,611	134,361	139,854	136,733
Average Manual Premium	\$8,160,533	\$8,365,739	\$7,844,963	\$7,531,095	\$7,147,168	\$6,677,185	\$6,021,710
Average Security Deposit	\$7,021,307	\$7,647,415	\$7,547,495	\$6,575,029	\$6,833,097	\$6,379,221	\$6,363,226

Notes: Panel (a) displays the percentage of all reported non-subscriber injuries (which, in raw form, makes up the final row) made up of claims along various dimensions (i.e., rows). Panel (b) displays the total number of companies, establishments, and employees among self-insured employers, along with average manual premiums and security deposits per firm (indexed to 2021 U.S. dollars). Source: Texas open records requests, 2015-2021.

The third and final component is annual reports of all self-insured employers in Texas, who are also subscribers to the workers' compensation system. These data, spanning 2009-2025, report employer names, industries, counts of establishments and employees, security deposit sizes (and types, in more recent years, such as via cash or surety bonds), and manual premiums. Becoming self-insured requires government certification, with rules that differ by

¹³I present the data as it was provided, which broke White/Black/Hispanic into mutually-exclusive categories. These are the first estimates published of their kind; no comparable data exists in the subscriber administrative data.

state. As such, becoming self-insured is presumably costly, which explains why - since 2016 - no employers have created new self-insured arrangements.

Summary information about self-insured employers is reported in panel (b) of Table 1. Over time, fewer and fewer firms decide to be self-insured, which also cuts down the number of establishments. The number of employees at self-insured firms decreases by 6% over the 2015-2021 period, which is much smaller than the roughly 25% reduction in the number of firms, implying the exiting firms were, on average, smaller. Reductions in the average manual premium, amounting to a reduction of 26% over the 2015-2021 period, reflect decreases in loss costs over time. These premiums are not paid to any insurer and only estimate a baseline premium to compare to privately-insured firms. Notably, reported security deposit amounts did not change nearly as much as average manual premiums, with a 9% reduction over the 2015-2021 period. This friction between cuts in perceived costs of injuries (manual premiums) and an administrative cost of being self-insured (security deposits) may suggest one reason why fewer firms decide to remain self-insured over time.

3.2 Other Data

To build out labor market information at the industry level, the QCEW provides information related to the total number of establishments, total amount of payroll (to create an uncovered rate, for example), and average weekly and annual wages that adjust for the number of employees and average work time.

4 Theoretical Framework

What pushes firms to opt in or out of the workers' compensation system? This is the guiding question of this analysis, and it is unlikely to be completely answerable in one dimension. Instead, this paper posits two key reasons as being related to the cost of workers' compensation insurance and the ability for firms opting out to avoid full wage pass-through to workers. This section describes the framework under which employers make this coverage decision and motivates the forthcoming two sections on prices and wage pass-through.

4.1 Opting In

The cost of opting in to the workers' compensation system is relatively straightforward for an employer. An employer either pays a premium for workers' compensation coverage or pays administrative costs (and chooses to bear risk) to set up a certified self-insured arrangement. In exchange for providing either of these plans to their employees, the employer minimizes their legal liability risk arising out of tort suits from injured employees.

When a firm chooses to purchase workers' compensation insurance coverage, they are subject to a pricing scheme that is quite complex, including advanced levels of experience rating that will differ with firm size. Many elements of the lengthy workers' compensation rate algorithm only apply to certain firms. Thus, in the forthcoming analysis, I analyze just the first element, but an element - nonetheless - that applies to all employers; this element is referred to as the manual premium. The manual premium is given by:

$$ManualPremium_{o,f,i,t} = ManualRate_{o,i,t} * Payroll_{o,f,t}, \quad (1)$$

for an occupation o in firm f buying from insurer i in policy year t . The second term, $Payroll_{o,f,t}$ is the amount of payroll, in dollars, a given firm employs in a certain occupation and policy year. The first term is provided by:

$$ManualRate_{o,i,t} = LossCost_{o,t} * Loadings_{o,i,t}. \quad (2)$$

In this setup, $LossCost_{o,t}$ represents a bureau rate per \$100 of payroll that is applied to all employers of a given occupation in a certain policy year. This rate is set by the National Council on Compensation Insurance (NCCI) and reflects expected costs for paying losses and loss adjustment expenses. These rates are typically updated on July 1 of each year, which defines a policy year. The second component, $Loadings_{o,i,t}$, is an insurer-specific component that covers insurer costs beyond losses and loss adjustment expenses, such as for profit margins and operating expenses. It also includes possible risk charges, which adjust

the premium upward for potential variation in loss expectations. In other words, there is a component of insurer loadings that allows for increasing or decreasing the premium to reflect disagreement with the loss cost levied by the NCCI.

With respect to the overall rate for coverage, we expect downward-sloping demand (Cabral et al., 2022). Yet, there are reasons to believe that firms may respond to each element of Equation 2 differently. The loss cost component reflects both a change in price and in insurance value, given loss costs represent loss payments expected based on past experience. Firms may be more sensitive to loss costs, rather than loadings, if they believe a change in loss costs is too large relative to the perceived change in insurance value. On the other hand, firms may be more responsive to loadings if they reflect profit margins in an already-profitable market or if loadings help to adjust premiums up or down in response to disagreement with loss costs. Given Cabral et al. (2022) estimate demand sensitivity to changes in loss costs, I contribute to the literature by estimating demand sensitivity for loss costs and loadings separately. In the forthcoming sections, I will also document patterns in workers' compensation insurer loadings for the first time (to my knowledge).

Self-insured employers, all else equal, face a similar expected loss to if they had purchased insurance, given benefits are prescribed by statute. However, self-insured firms can escape the $Loadings_{o,i,t}$ component, particularly the insurer profit margin and operating expense components. Self-insured employers do, though, carry some level of variation in losses themselves, given they are retaining risk with respect to claims. This level of variation ought to decrease in firm size. They may also purchase stop-loss coverage, more or less reinsuring their own self-insurance arrangement, and will need to accept administrative costs associated with setup (including the costs of arranging security deposits). Given there are no newly self-insured firms after 2016 in Texas, I largely focus on the decision between purchasing workers' compensation to opt in or opting out.

4.2 Opting Out

Opting out means avoiding workers' compensation insurance premiums. As discussed above, these plans are often replaced with alternative injury plans that are, on average, about 50% less generous (Jinks et al., 2020). Opting out of the workers' compensation system, though, also opens firms up to legal liability risk from workplace injuries and illnesses. This may suggest only firms in low-injury industries ought to opt out. In the face of restricted common law defenses (i.e., the fellow servant rule, assumption of risk, and contributory negligence), firms may perceive liability risk to be daunting if employees are cognizant of their legal recourse options.

While there exist no estimates as to the expected cost of tort suits from non-subscribed employers (even if data on damage awards or settlements were available, calculating the probability of lawsuit would be difficult), one way for an employer of any risk (injury) level to be willing to accept such risk is to be able to remove formal workers' compensation coverage and rely on the lack of salience among workers. This lack of salience could be with respect to the chance of suing in court or in the negotiation for increased wages in the face of greater risk. In the analysis that follows, I provide, to my knowledge, the first estimates of wage pass-through when workers' compensation benefits are removed at the extensive margin, building on work that shows wages increase when the programs were created (Fishback and Kantor, 1995) or marginal benefit increases were legislated (Gruber and Krueger, 1991).

4.3 Motivations and Injury Levels

With this motivation, a firm decides to opt out of the system if:

$$P > [\gamma * L] + E + [W * \theta], \quad (3)$$

where P represents the workers' compensation premium, driven by - for all firms - two key, multiplicative elements: loss costs and loadings. On the right-hand side, expected tort costs come from the product of γ , which is a multiplicative factor of the chance of injury and

the subsequent chance of lawsuit (or settlement), and the eventual damage award, L . E represents the cost of an alternative injury plan (if provided). Finally, removing formal coverage and replacing it with either 1) no coverage or 2) a less generous alternative injury plan ought to generate an increase in wages, W . θ adjusts the wage increase by an estimate of pass-through, which depends on the salience of coverage and the outside option (which also affects γ through the chance of filing suit, conditional on injury or illness). Given that γ , L , and E are not observable or estimable in this context, I use the unique set of administrative data available to study how $[W * \theta]$ and two key elements of P - loadings and loss costs - drive the decision to opt out of the system.

Plausibly, the effects of each element may differ by the risk level of an industry. Beyond providing results as to the rationales above generally, I also study how they vary by injury levels across industries. It is ex-ante ambiguous whether firms in high- or low-risk industries will respond differently to each element of the premium, which critically depends on if, and how, loadings adjust loss cost factors. For example, if firms in low-risk industries perceive themselves to be overcharged with respect to loss costs, they may be more sensitive to the loadings component (where decreases in loadings help keep them in the system) than firms in high-risk industries. In this context, the government coordinates with a bureau to set loss costs and allows insurers to control the loadings space, so sensitivity estimates for each element may be valuable to policymakers in achieving desired levels of system participation in different industries.

It is also ex-ante ambiguous whether firms in high- and low-risk industries are able to use wage pass-through differently in opting out. If employees are fully-engaged and risk is salient, firms in high-risk industries ought to provide higher wage pass-through when removing coverage. On the other hand, if salience is limited, firms in high-risk industries may be able to use limited wage pass-through to justify opting out of the system. Unionization may be an important mediator in this context.¹⁴

¹⁴As a note for the reader: unionization is not yet included in this draft but the data have been collected.

5 Descriptive Evidence

The section below begins by describing the creation of covered rates in the data and the subsequent variation in said rates. Then, I provide a focus on injuries, within and outside of the covered system. This injury analysis takes a focus of how covered rates vary with the “safety” of an industry, which is derived from an industry’s injury rates.

5.1 Measuring Covered Rates

I make use of the extensive administrative data to construct the covered rate, which is defined as:

$$CoveredRate_{n,t} = \frac{CoveredPayroll_{n,t}}{TotalPayroll_{n,t}}, \quad (4)$$

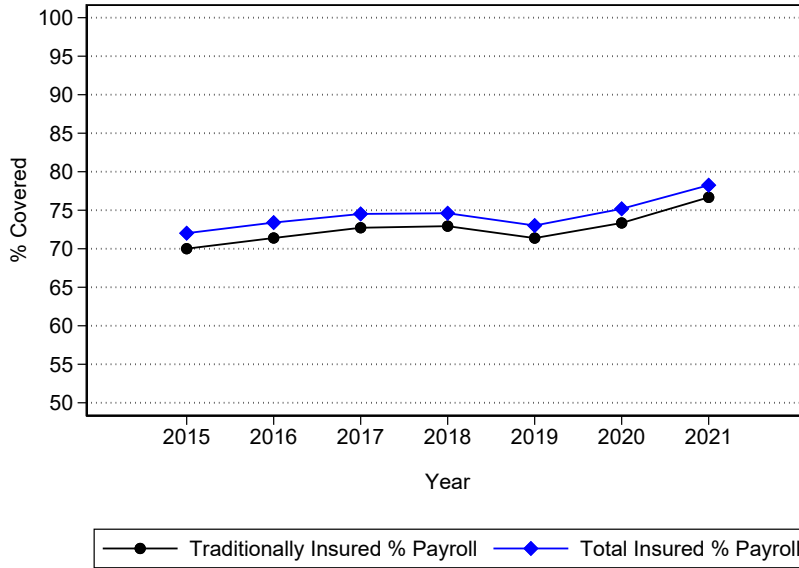
where the numerator is the sum of payroll covered in industry n and year t by privately-insured and self-insured employers. These inputs come from the administrative data. The denominator comes from the QCEW.¹⁵ Figure 1 below displays statewide covered rates over the analysis period. In general, overall covered rates have increased over time, as given by the blue (diamond) line, which combines privately-insured and self-insured payroll. In 2015, 72% of payroll was covered by the workers’ compensation system; by 2021, 78% of payroll was covered. The presence of self-insured payroll is relatively stable over time, as given by the gap between the two lines. Appendix Figure A.1 shows how these constructed rates compare to the only public data available on covered rates. These public estimates come from a TDI survey of employers (DWC, 2022). In general, the constructed covered rate mirrors those estimates well but are slightly different, given TDI measures covered rates as the percentage of employees and employers but not payroll.¹⁶

One challenge in this setting is defining the covered rate at the industry level, because

¹⁵A larger appendix on the precise construction of this measure is being written; I have tried, for the purpose of the conference, to describe it below for feedback purposes.

¹⁶There is an additional way to cut these data using zip-code information, which is by county. I do not currently present this in the main text because I have not settled on how or if to use county-level information. I have added Appendix Figures A.2 and A.3 to highlight this variation. I would be delighted to get feedback on ideas for using this variation.

Figure 1: Covered Rates Over Time



Notes: This figure displays covered rates over time, with and without self-insured employers included. Sources: Texas open records requests and QCEW, 2015-2021.

privately-insured payroll is provided at the occupation level, while self-insured estimates and the QCEW’s universe of payroll are reported at the industry level using the North American Industry Classification System (NAICS). To crosswalk between occupations (which use codes provided by NCCI and are specific to Texas employers) and industries, I primarily use an industry cross-reference tool from the International Risk Management Institute (IRMI). This maps direct matches, which applies to occupations that are specific enough to yield one industry match. For several larger occupations that are employed in many industries - such as clerks, salespeople, executives, and officeworkers - I use the BLS’ industry-occupation matrix for each year. This matrix provides the percentage of a given occupation that is employed in a given industry. The estimates are national, so I have to assume that the within-occupation industry distributions of employment are similar in Texas as they are nationally.

To improve the precision of one-to-one matches using the IRMI tool, I present most of my baseline estimates using the three-digit NAICS code. Using more granular codes erodes

one-to-one matching and relies more on the BLS matrix, and it also leads to under-counting of the denominator of Equation 4 because BLS does not report payroll in certain industries in Texas at more granular levels. While payroll estimates are provided directly by the privately-insured administrative data and via the QCEW, the self-insured data include only the number of employees. To turn that into payroll, I multiply the employee counts by the QCEW’s reported average annual wage within that industry and year. The self-insured data reports NAICS codes, so no crosswalk is needed to the QCEW.

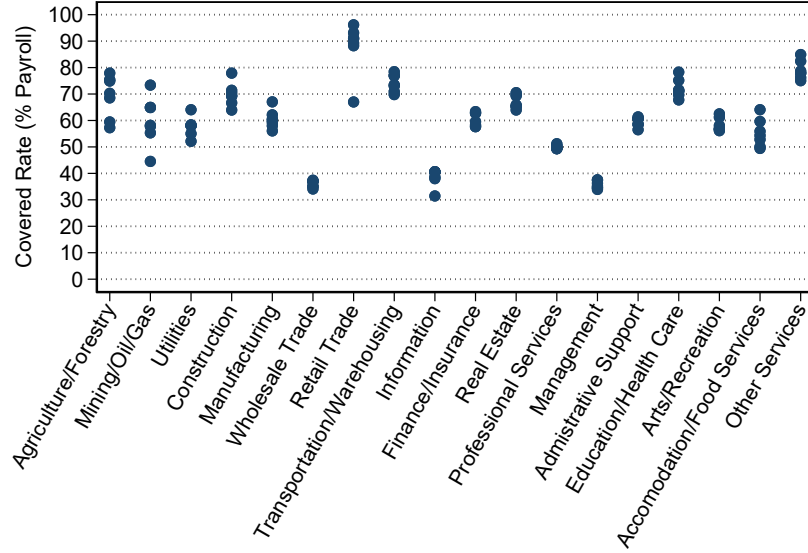
Figure 2 depicts variation in covered rates across industries and over time, binning industries by two-digit NAICS code categories, where each dot represents the covered rate within a given industry in each year. I use the two-digit code for this exercise because it is the most complete picture of payroll counts in the QCEW. There is variation across industries and over time. In some industries, such as wholesale trade, information, and management, covered rates are always below 50%. In other industries, such as retail trade, education, health care, and transportation, covered rates are always above 70%. Yet, in other industries the covered rate changes significantly over time, especially in agriculture, mining, and oil/gas extraction.

5.2 Covered Rates and Industry Injury Rates

As discussed in Section 4, it is plausible that firms in high- and low-risk industries may respond to the coverage choice in different ways. One component of that decision was the expected cost of tort suits, which critically relates to the restrictions in employer defenses against injury claims when opting out. Thus, one question that has not yet been answered in this space: do firms in unsafe industries opt out to begin with? If firms are only focused on the expected tort costs and perceive employees as being fully knowledgeable of injury risk and tort options, we may not expect firms in unsafe industries to opt out because both the probability of injury (and eventual suit) and expected payouts may be higher (i.e., because of potential punitive damages).

Figure 3 presents binned scatterplots, binning industries by a measure of safety (using,

Figure 2: Covered Rates by Industry Over Time

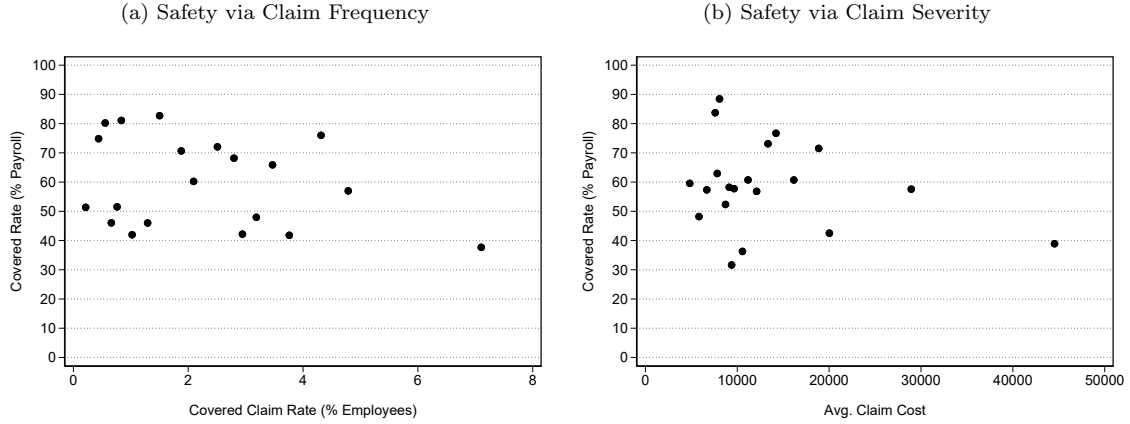


Notes: This figure displays covered rates on the vertical axis within two-digit NAICS code categories on the horizontal axis. Each dot within an industry represents a year. Sources: Texas open records requests and QCEW, 2015-2021.

in panel(a), the claim rate among opt-in firms, from the administrative data; panel (b) uses the average claim cost among opt-in firms, excluding zeroes) via the horizontal axis, and then reading off that bin's covered rate on the vertical axis. While these figures depict industries at the three-digit NAICS level, Appendix Figures A.4 and A.5 do the same for the two- and four-digit levels. The patterns are similar, with the caveat that changing industry granularity affects how much payroll QCEW reports without censoring (i.e., covered rates shift downward at the four-digit level given some industries are too small for QCEW to safely report payroll for).

The patterns in these figures indicate that firms in less safe industries do, in fact, opt out of the system. In the least safe bucket of industries with respect to claim frequency in panel (a), where roughly 7% of employees have a workers' compensation claim in a given year, the covered rate is less than 40% of payroll. When safety is measured in terms of claim severity, panel (b) indicates that the pattern is also persistent there - only about 40% of firms choose to be in the system while operating in industries with average claim sizes exceeding \$40,000.

Figure 3: Covered Rates and Industry Safety



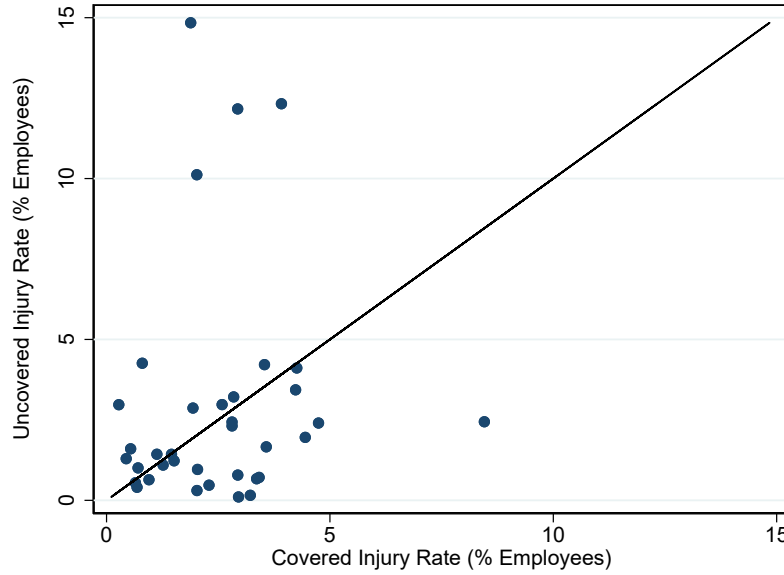
Notes: These figures display binned scatterplots of covered rates within 20 equally-sized bins of industries based on their covered claim rates in panel (a) and covered claim costs in panel (b). Industry is measured at the three-digit NAICS level. Sources: Texas open records requests and QCEW, 2015-2021.

How do injury rates vary, within the same industry, between firms opting in and opting out? Figure 4 provides an answer. There is not full reporting in all industries among the uncovered firms, so I provide caution that there is likely to be under-reporting (and not over-reporting) of injuries among uncovered firms. Nonetheless, graphing this against the 45-degree line suggests whether opting in (covered) or opting out (uncovered) firms are systematically different injury-wise within an industry.

In the direction of potential advantageous selection, north of the 45-degree line, roughly half of industries reported having higher uncovered injury rates than covered injury rates. There are several extreme cases, where the injury rate among uncovered firms more than triples the injury rate among covered firms. There are not as many extreme examples in the direction of adverse selection, south of the 45-degree line, but roughly half of industries have higher injury rates among opting-in firms than in opting-out firms.

The patterns in this series of graphs suggest that firms in unsafe industries do choose to opt out and that, even in the same industry, there is some evidence that opt-out firms are less safe than opt-in firms. This motivates not only a study of why a firm may opt out but also how the motivations to opt out may differ by industry safety.

Figure 4: Injury Rates, Opting In vs. Opting Out



Notes: This figure displays a scatterplot of injury rates at the level of industries, with covered firms on the horizontal axis and uncovered firms on the vertical axis. Sources: Texas open records requests and QCEW, 2015-2021.

6 Price Sensitivity

This section has two main goals: 1) document how firms respond to different elements of the workers' compensation premium by opting in or out, and 2) provide estimates of how these demand sensitivities may vary by industry risk level. By documenting baseline demand sensitivity to various premium elements and how that varies with injury levels, these results ultimately can assist policymakers looking to pull premium levers to achieve optimal coverage rates (in Texas, where coverage is optional) and in considering removing coverage mandates altogether.

This section makes use of the administrative data for firms opting-in and does not rely at all on an industry crosswalk. By way of providing summary statistics for these data, which can now be properly tied to injury-levels and rates, Table 2 introduces the data.¹⁷ The overall sample is provided in column (1), which is at the occupation-firm-year level (i.e., the

¹⁷In the administrative data, manual rates are provided, which are the product of loss costs and loadings. To decompose these values, loss cost values were taken from the TDI website, provided for each year. The loss costs are available here: <https://www.tdi.texas.gov/wc/regulation/rcomp.html>.

level at which workers' compensation insurance is priced). The remaining columns split the sample below and above the median risk level, defined by having a non-zero injury count in columns (2) and (3), and a non-zero total claim cost in columns (4) and (5).

Table 2: Summary Statistics

	(1) Overall	(2) Above Median Risk - Frequency	(3) Below Median Risk - Frequency	(4) Above Median Risk - Severity	(5) Below Median Risk - Severity
Payroll (\$100s)	9,191.11 (134,181.20)	56,403.15 (420,652.20)	4,732.18 (52,563.10)	62,293.50 (442,318.80)	4,768.31 (54,358.40)
Claim Count	0.31 (5.54)	3.56 (18.55)	0.00 (0.00)	3.97 (19.61)	0.00 (0.06)
Average Claim Cost (\$)	1,102.02 (28,169.10)	12,737.74 (94,894.00)	0.00 (0.00)	14,333.44 (100,653.10)	0.00 (0.00)
Claim Count <i>firm level</i>	1.30 (20.15)	15.12 (67.04)	0.00 (0.00)	16.67 (70.22)	0.02 (2.77)
Average Claim Cost (\$) <i>firm level</i>	1,094.38 (26,421.60)	12,681.95 (89,122.60)	0.00 (0.00)	14,175.50 (94,279.30)	4.88 (704.60)
Total Premium (\$)	8,959.01 (111,508.70)	63,326.89 (363,854.00)	3,824.25 (28,276.10)	70,651.22 (384,680.60)	3,820.78 (28,311.80)
Loss Costs	0.74 (0.99)	1.31 (1.11)	0.69 (0.96)	1.35 (1.12)	0.69 (0.96)
Loadings	2.25 (1.24)	2.11 (1.03)	2.27 (1.25)	2.16 (1.06)	2.26 (1.25)
Experience Rate	0.49 (0.50)	0.78 (0.48)	0.47 (0.50)	0.82 (0.46)	0.47 (0.50)
Occupation-Firm-Year Observations	3,422,509	295,344	3,127,165	263,138	3,159,371

Notes: This table provides summary statistics for the baseline analysis sample, at the level of occupation-firm-year. Standard deviations are provided in parentheses. Source: Texas open records requests, 2015-2021.

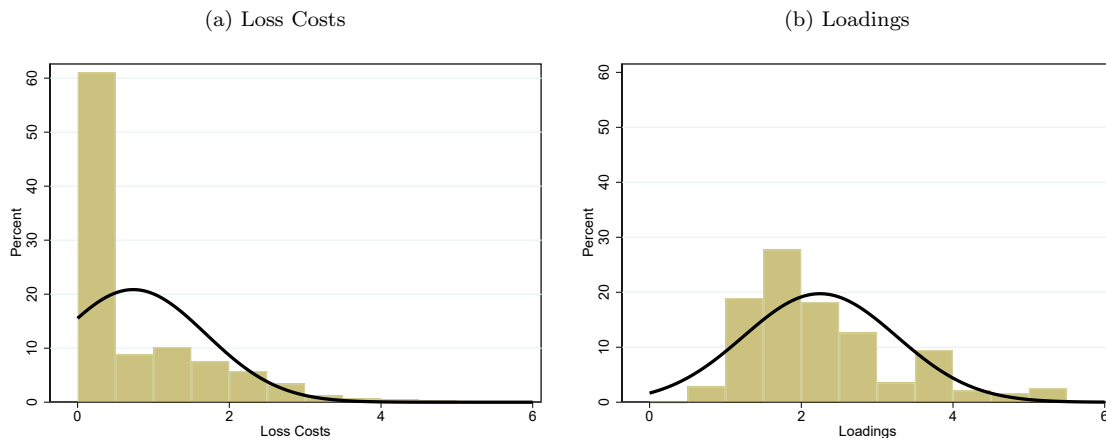
The table suggests that most occupation groups at firms do not experience claims. In addition, the no-claim classification-firm-year observations tend to employ fewer people. The average occupation group within a firm experiences four claims per year, conditional on being a non-zero firm with respect to claims. The average severity is more than \$10,000 per claim. As a sense-check in the data, loss costs are higher among higher risk occupation-firm-year observations, as are experience rates. This, combined with higher payroll amounts for high risk firms, makes premiums much larger. Interestingly, the opposite is true for loadings,

which could be attributed to insurers disagreeing with loss cost assessments provided by NCCI.

6.1 Variation in Loss Costs and Loadings

Section 4 described the differences between loadings and loss costs, which are two elements that affect every firm’s workers’ compensation premiums and, thus, the baseline against which firms compare the benefits of opting out. Figure 5 documents baseline variation in each rate element, taken from the data at the level of occupation-firm-year. Loss costs, in panel (a), are expressed as a dollar charge per \$100 of annual payroll, suggesting most observations experience a loss cost charge of \$0.50 or less per \$100 of payroll. This is consistent with many firms employing individuals in the most common and most safe occupations, such as executives, office workers, and salespeople. The distribution is right-skewed, indicating some occupations receive much higher loss cost charges.

Figure 5: Variation in Loss Costs and Loadings



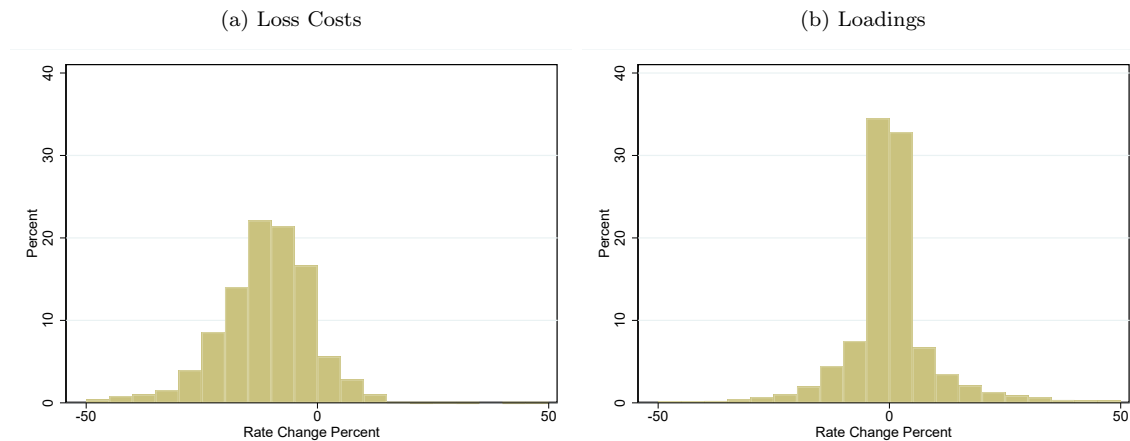
Notes: These figures display loss cost and loadings rates in panels (a) and (b), respectively. The histograms break rates into 0.5-width buckets and display the percent of occupation-firm-year observations with such a rate. Sources: Texas open records requests, 2015-2021.

On the other hand, in panel (b), loadings charges take a more normal distribution. These ought to be interpreted as multipliers applied to loss cost rates. For example, if a given occupation has a loss cost of \$0.75 per \$100 of payroll and a loadings value of 2, then the insurer is charging \$1.50 per \$100 of payroll (or roughly double the expected value

of losses). Given loadings represent an insurer's need for a profit margins and to recoup operating expenses, loadings are rarely below 1.

Figure 6 shows the distribution of year-over-year updates to loss costs and loadings values. Over the course of the sample period, most loss cost updates are negative, indicating that loss costs are being systematically decreased over time. On the other hand, loadings updates are more evenly distributed, though smaller in absolute terms than loss cost updates. This further supports the notion that loss costs and loadings are picking up on different patterns and, thus, may alter demand sensitivity (opt-out sensitivity) differently. When I regress loadings updates on loss cost updates, the resulting coefficient is -0.51 with a standard error of 0.02, suggesting that when loss costs change in one direction, loadings change roughly half as much - on average - in the opposite direction.

Figure 6: Variation in Updates to Loss Costs and Loadings

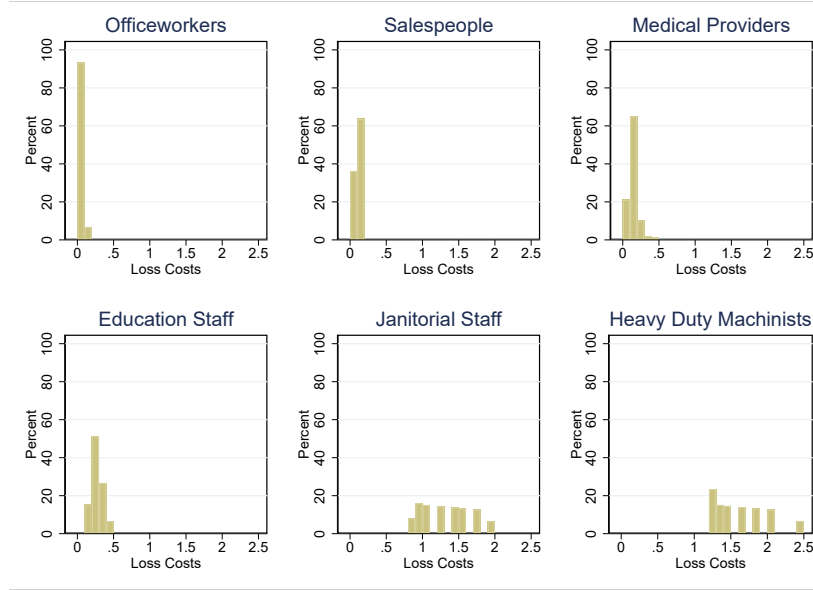


Notes: These figures display year-over-year percentage changes in loss cost and loadings rates in panels (a) and (b), respectively. The histograms break changes into 5%-width buckets and display the percent of occupation-firm-year observations with such a change. Sources: Texas open records requests, 2015-2021.

Those differing patterns are, as Table 2 suggested, in part related to risk levels. After all, loss costs directly represent occupation risk levels via expected losses. Figures 7 and 8 support this assertion further, showing the distributions of loss costs and loadings, respectively, across six common occupation categories. These categories include officeworkers, salespeople, medical providers, education staff, janitorial staff, and heavy duty machinists.

In the safer occupations (officeworkers, salespeople, medical providers, and education staff) the distribution of loss costs is much different (and less skewed) than it is in less safe occupations (janitorial staff and heavy duty machinists). The same cannot be said for loadings, where the distributions are all relatively consistent.

Figure 7: Loss Cost Distributions by Occupation



Notes: This figure displays loss cost distributions within each occupation category. Sources: Texas open records requests, 2015-2021.

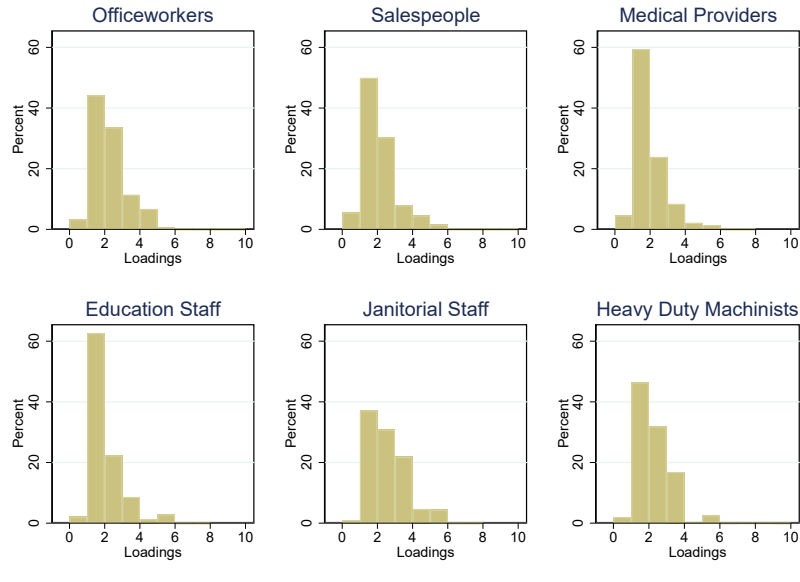
6.2 Demand Sensitivity Analyses

To study the sensitivity of demand to changes in loss costs and loadings, I run regressions of the following design, which replicates the methods of [Cabral et al. \(2022\)](#):

$$\ln(\text{Payroll}_{o,t}) = \alpha + \theta \ln(\text{Loadings}_{o,t}) + \beta \ln(\text{LossCosts}_{o,t}) + \gamma_o + \delta_t + \epsilon_{o,t}, \quad (5)$$

where the outcome is covered payroll within an occupation, o , in policy-year t . The specification includes occupation and policy-year fixed effects to identify off of changes in loss costs and loadings within an occupation over time. While this design averages over loadings within an occupation, given they are created at the insurer level, this is a replication of the

Figure 8: Loadings Distributions by Occupation



Notes: This figure displays loadings distributions within each occupation category. Sources: Texas open records requests, 2015-2021.

Cabral et al. (2022) method using similar Texas data (though over a different time period).¹⁸ Standard errors are clustered at the occupation level. In this setup, we can interpret changes in covered payroll as changes in coverage at the extensive margin because 1) no new firms become self-insured over this time period and 2) firms cannot manipulate their occupation code from year-to-year. It is not possible, however, to rule out that exits are firm failures.

Table 3 below provides the baseline results. Column (1) is a pure replication of Cabral et al. (2022), though over a different set of years and a slightly longer time period. In their analysis, the comparable coefficient was -0.293 with a standard error of 0.122. My estimate is smaller and not as precisely estimated relative to their result, but the results are broadly similar. Running the same estimation, in column (2), only with loadings suggests firms are more sensitive to loadings than to loss costs. When running them both together

¹⁸If readers are interested in running this regression at the insurer level, Appendix Table A.1 provides those estimates. In those results, loadings appear to have even more demand sensitivity, given there is more variation at the insurer level. The challenge with that specification, though, is that the administrative data do not track firms across insurers, so covered payroll at the insurer level as an outcome suggests a mixing of the intensive (changing insurers) and extensive (opting out of the system) margins.

in the same estimation, in column (3), firms still appear more sensitive to loadings than to loss costs. Comparing the coefficients from the first two columns to the third, interestingly, supports the discussion above about movement in opposite directions; specifically, it appears not accounting for loadings attenuates the estimate of loss costs’ effect. This suggests that loadings and loss costs are negatively related in terms of changes within an occupation.

Table 3: Baseline Insurance Demand Estimates

	(1) ln(Covered Payroll)	(2) ln(Covered Payroll)	(3) ln(Covered Payroll)
ln(Loss Costs)	-0.212** (0.103)		-0.377*** (0.121)
ln(Loadings)		-0.466** (0.201)	-0.641*** (0.216)
Classification Fixed Effects	Yes	Yes	Yes
Policy-Year Fixed Effects	Yes	Yes	Yes
Observations	2,876	2,876	2,876

Notes: This table provides results for estimating regressions of Equation 5’s design. Standard errors are clustered at the classification level. Source: Texas open records requests, 2015-2021. Statistical significance is given by: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

How do the demand sensitivity results vary by the injury levels of an occupation? Table 4 duplicates the baseline demand estimation, with estimates separately for occupations that have total costs (i.e., the number of injuries multiplied by the average claim severity) above the median total costs among all occupations in a given year. Columns (1)-(3) show results for occupations below the median cost level, while columns (4)-(6) apply to occupations above the median. These results suggest that firms employing people in low injury cost occupations are very responsive to changes in both loss costs and loadings, though much more so to loadings. Firms employing people in high injury cost occupations are not responsive to changes in prices.

One natural next step is to ask if the strong demand sensitivity differences by industry injury costs is related to how over- or under-charged a given occupation with respect to loss costs, and whether loadings make up for that. To test for this, I create a “homemade” loss cost estimate that is observed losses in the administrative data per \$100 of covered

Table 4: Demand Estimates Relative to Median Costs

	Below Median Cost			Above Median Cost		
	(1)	(2)	(3)	(4)	(5)	(6)
	ln(Covered Payroll)	ln(Covered Payroll)	ln(Covered Payroll)	ln(Covered Payroll)	ln(Covered Payroll)	ln(Covered Payroll)
ln(Loss Costs)	-0.237 (0.158)		-0.415** (0.166)	-0.017 (0.064)		-0.026 (0.095)
ln(Loadings)		-0.695*** (0.249)	-0.839*** (0.258)		-0.005 (0.086)	-0.024 (0.130)
Classification Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Policy-Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,380	1,380	1,380	1,440	1,440	1,440

Notes: This table provides results for estimating regressions of Equation 5’s design, separately for occupations below median cost levels in columns (1)-(3) and above median cost levels in columns (4)-(6). Standard errors are clustered at the classification level. Source: Texas open records requests, 2015-2021. Statistical significance is given by: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

payroll, all at the occupation-year level. I measure observed losses as the sum of losses incurred (both medical and indemnity), loss adjustment expenses incurred, and employer attorney fees incurred - all measured at 3.5 years after injury date (or sooner if the case was closed sooner), which is consistent with Cabral et al. (2022). Then, I estimate Equation 5, focused only on loadings, separately for firms with positive and negative loss cost gaps, where $LossCostGap_{o,t} = LossCostActual_{o,t} - LossCostHomemade_{o,t}$. Thus, a positive (negative) loss cost gap indicates overcharging (undercharging).

Table 5 presents the results. The results here suggest that loadings changes generate significantly larger demand responses when they are made for occupations where employers perceive being overcharged. This is consistent with the descriptive findings above, and given that being in a low injury cost occupation is significantly, positively related to a positive loss cost gap, the demand estimates presented above can be explained in part by the fact that loadings help keep employers with low injury cost occupations in the system.

On the high injury cost occupation front, we know this does not mean that these firms are always purchasing coverage, given the descriptive results from Section 5. Thus, firms employing people in these occupations must be choosing to opt out for reasons other than insurance premiums. This could be true because firms are strongly experience-rated in workers’ compensation or because changes in loss costs reflect real changes in the cost of

Table 5: Loss Cost Gaps and Loadings

	Gap = -	Gap = +
	(1)	(2)
	ln(Covered Payroll)	ln(Covered Payroll)
ln(Loadings)	-0.151 (0.146)	-0.671** (0.314)
Classification Fixed Effects	Yes	Yes
Policy-Year Fixed Effects	Yes	Yes
Observations	1,118	1,665

Notes: This table provides results for estimating regressions of Equation 5's design, but only with loadings. The regressions are estimated separately depending on whether the loss cost gap (as defined in the text) is negative, as in column (1), or positive, as in column (2). Standard errors are clustered at the classification level. Source: Texas open records requests, 2015-2021. Statistical significance is given by: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

injuries, shifting the insurance value in line with price (leaving little shock to demand and little motivation for loadings to adjust due to insurer disagreement with loss costs). One reason other than insurance premiums to opt out of the system is related to wage pass-through, which is studied in the next section.

7 Wage Pass-Through

Beyond the cost of insurance, another determinant of opt out behavior could be the extent to which employers opting out of the workers' compensation system can avoid complete pass-through of newly-shifted risk to employee wages. This is tied to employee salience in this setting, which is a potentially policy-relevant friction. This section first estimates how covered rates in an industry affect wages, and then uses that estimate as a component of a larger wage pass-through computation. The differences in wage pass-through are then discussed by industry injury levels.¹⁹

¹⁹This section studies wages, which are reported via the QCEW. Thus, covered rates are the independent variable of interest, which requires cross-walking as described earlier. I realize that endogeneity likely plays a role in this portion of the paper, and I would appreciate any feedback - especially on the potential usage of instrumental variables (such as judge elections for the courts that oversee non-subscriber injury cases) to better isolate the effect of covered rates on wages. This is all still very much in development.

7.1 Covered Rates and Wages

One element of the pass-through of losing workers' compensation insurance coverage into wages is understanding how changes in an industry's covered rates relate to changes in an industry's average annual wage. Thus, I estimate the following specifications:

$$\ln(AnnualWages_{n,t}) = \alpha + \beta \ln(CoveredRate_{n,t}) + \gamma_n + \delta_t + \epsilon_{n,t}, \quad (6)$$

where $AnnualWages_{n,t}$ represents average annual wages in an industry, n , in year t . This element comes from the QCEW. The $CoveredRate_{n,t}$ term is the same as defined in Equation 4, and the specification includes industry and year fixed effects. Clustering of standard errors is done at the industry level. Column (1) of Table 6 provides the baseline results. The estimate suggests that decreasing the covered rate by 1% is related to a 0.07% increase in average annual wages.

Table 6: Covered Rates and Wages

	Overall	Above Median Injury Rate	Below Median Injury Rate
	(1)	(2)	(3)
	$\ln(\text{Average Annual Wage})$	$\ln(\text{Average Annual Wage})$	$\ln(\text{Average Annual Wage})$
$\ln(\text{Covered Rate})$	-0.067** (0.032)	-0.008 (0.023)	-0.133*** (0.048)
Industry Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Observations	574	285	285

Notes: This table provides results for estimating regressions of Equation 6's design. Standard errors are clustered at the industry level. Columns (2) and (3) divide industries into being above or below median industry injury levels, respectively. Source: Texas open records requests, SOII, and QCEW, 2015-2021. Statistical significance is given by: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Columns (2) and (3) focus on whether or not this responsiveness of wages to covered rates varies with industry injury rates. Given this analysis is cross-walked to the three-digit NAICS code level, I merge in data from the SOII on injury incidence rates and divide the sample above and below the median. The results suggest that industries with higher injury rates see very little response in wages from changes in the covered rate. In low injury rate

industries, the opposite is true; there is a statistically significant, negative response in wages from changes in the covered rate.

7.2 Defining Wage Pass-Through

The estimates above do not tell us about the prevalence of wage pass-through directly. As a result, I propose the following pass-through²⁰ measure in this context:

$$PassThrough_{n,t} = \frac{WageChangePredicted_n}{CostsShifted_{n,t}}, \quad (7)$$

where the numerator is the product of 1) the β coefficient on from running industry-specific regressions of the form in Equation 6 (though with only year fixed effects) and 2) 1% of total wages in a given industry and year. The denominator is derived from the product of: 1) 1% of covered payroll (in \$100s) in a given industry and year and 2) the average loss cost in that industry and year. The loss cost reflects expected workers' compensation costs and, thus, expected costs shifted to employees by being out of the system. Multiplying it by 1% of covered payroll then estimates the total amount wages in a given industry and year should increase if pass-through is complete (i.e., $PassThrough_{n,t} = 1$).

One complicating factor in this context is whether or not complete wage pass-through really means that $PassThrough_{n,t} = 1$. It should be something less than 1, given we know that roughly 75% of employees working at opt-out firms have alternative injury plans that are about 50% as generous as workers' compensation programs (Jinks et al., 2020). Thus, $PassThrough_{n,t} = 1$ would be complete only for 25% of employees; for the other 75%, complete pass-through would mean $PassThrough_{n,t} = 0.5$. Therefore, complete pass-through on average - across all industries - would be more like 0.65.²¹

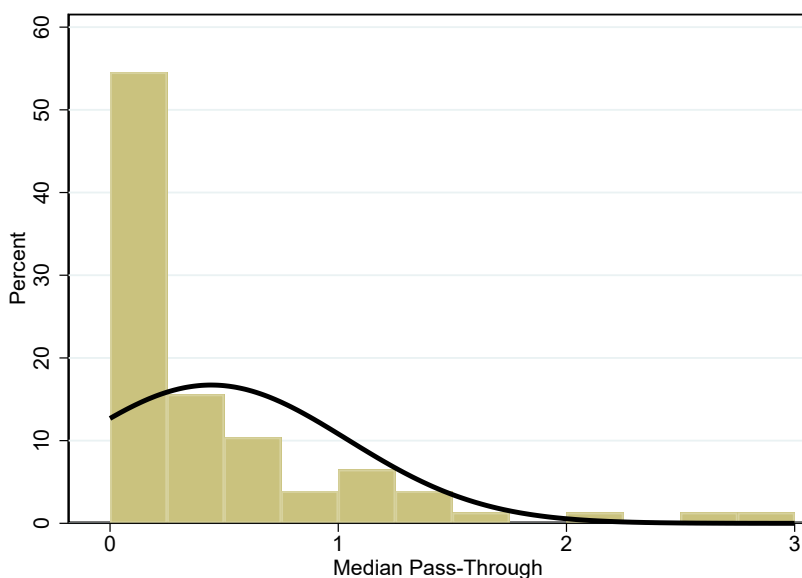
²⁰I realize that the numerator does not vary by time, but I believe this is a data constraint in that I cannot estimate coefficients from Equation 6 separately for each industry in each year. I am hoping for ideas on ways to allow the numerator to vary over time.

²¹This omits the role that lawsuits play. If the lawsuit option is salient, used efficiently, and without substantial time costs, it may reduce the level of pass-through that is needed to be "complete" even further than 0.65.

7.3 Estimating Pass-Through

Figure 9 depicts a histogram of median pass-through at the industry level. In general, most industries experience pass-through between 0 and 0.25, with roughly 70% experiencing pass-through less than 0.5, which is significantly below the complete level of 0.65. To get at variation over time, Figure 10 depicts median and mean pass-through levels in each year across all industries. Across all years, pass-through is less than complete statewide. In fact, median pass-through never exceeds 0.3. This is baseline evidence that incomplete wage pass-through of expected injury costs when opting out could be a motivation for firms.²²

Figure 9: Median Pass-Through

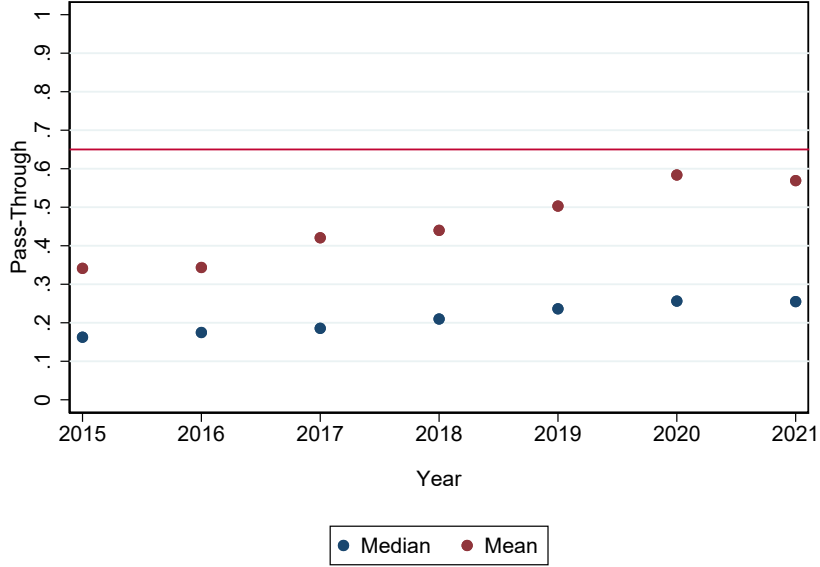


Notes: This figure displays a histogram for median pass-through at the industry level (i.e., three-digit NAICS level). Sources: Texas open records requests and QCEW, 2015-2021.

Next, Figure 11 depicts median and mean pass-through by year within industries that have injury rates (risk) below or above the statewide median. In this depiction, and consistent with the results from Table 6, industries that have injury rates below the statewide median have much larger pass-through values than industries with injury rates above the median. In fact, those below the median eventually get to complete pass-through over time, while

²²I expect that the upward trend in these measures over time comes from the lack of time variation in the coefficient on covered rates in the wage regression, hence footnote 20.

Figure 10: Pass-Through by Year



Notes: This figure displays median and mean pass-through by year. A horizontal line is drawn at 0.65 to indicate the estimate for complete pass-through. Sources: Texas open records requests and QCEW, 2015-2021.

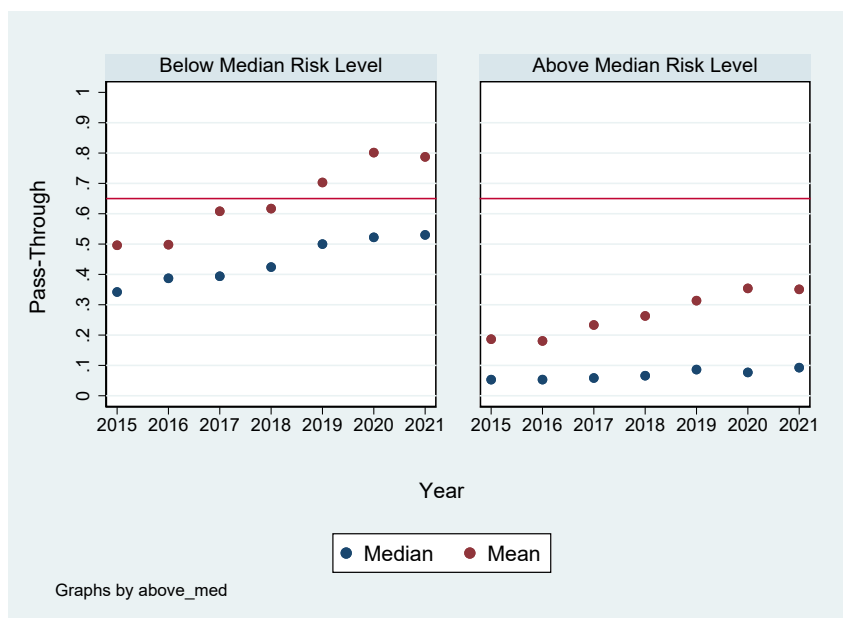
those above the median never see mean (median) pass-through exceed 0.4 (0.1). There could be many reasons for this. One that is untestable in this context is that workers in higher risk industries need less pass-through because their expected tort suit winnings are high. On the other hand, pass-through could be low because salience is low among employees (due to behavioral factors and/or the lack of presence of a union, which should increase salience of all employee benefits).²³

8 Discussion

In debates about the merits of optional coverage settings for workers' compensation benefits, reasons why employers decide to opt in or out of the system become relevant. To the best of my knowledge, very little is known in the current literature about how employers make this decision. To address this gap in the face of recent policy discussions in several states about repealing their coverage mandates, I study the one state in the US with a

²³As noted above, I plan to explore this unionization element further by understanding whether unionization mediates pass-through effects.

Figure 11: Pass-Through by Year, Industry Risk Levels



Notes: This figure displays median and mean pass-through by year among industries below or above median injury risk levels. A horizontal line is drawn at 0.65 to indicate the estimate for complete pass-through. Sources: Texas open records requests and QCEW, 2015-2021.

robust opt out: Texas. I make use of a unique administrative dataset to study how the cost of insurance and the ability to avoid complete wage pass-through when opting out drive firm decisions. I then study how those drivers vary with the injury levels of occupations and industries, which lends insight into which elements of premiums, for example, impact employers most relevantly.

First, I document descriptively that firms in unsafe industries do opt out of the workers' compensation system and that, within some industries, injury rates among opt-out firms are higher than for opt-in firms. Using these descriptive findings as motivation, I then largely replicate the findings of [Cabral et al. \(2022\)](#) and show that demand for being in the system is related to insurance prices. I build beyond that work, however, to show that this relationship is heterogeneous across occupation injury levels.

Namely, firms with employees in less safe occupations are much less price sensitive than firms with employees in safer occupations. For the first time, I provide a decomposition by premium elements, which shows that the firms with employees in safer occupations are

primarily pushed around the opt-out margin by the insurer-specific loadings charge. This charge allows insurers to adjust downward seemingly inflated loss cost estimates for these lower-risk occupations.

If firms in less safe industries are not pushed around the opt-out margin by insurance costs, what then matters to them? One plausible explanation is that they are better able to avoid increasing wages in lockstep with the risk-adjusted costs shifted to employees as they opt out. I show that, in fact, firms in less safe industries exhibit wage pass-through at levels half as large as what would be considered complete on average.

These results are relevant to policymakers in Texas specifically and across the US more broadly. In Texas, understanding which elements of premiums drive which types of employers to opt out is relevant, given each premium element is regulated differently. If regulators or policymakers have ideal opt-in rates, potentially even in certain industries or among certain occupations, then these findings isolate different policy levers for the first time. Broadly, the pass-through results motivate more study of how workers understand workers' compensation benefits (and their legal recourse options).

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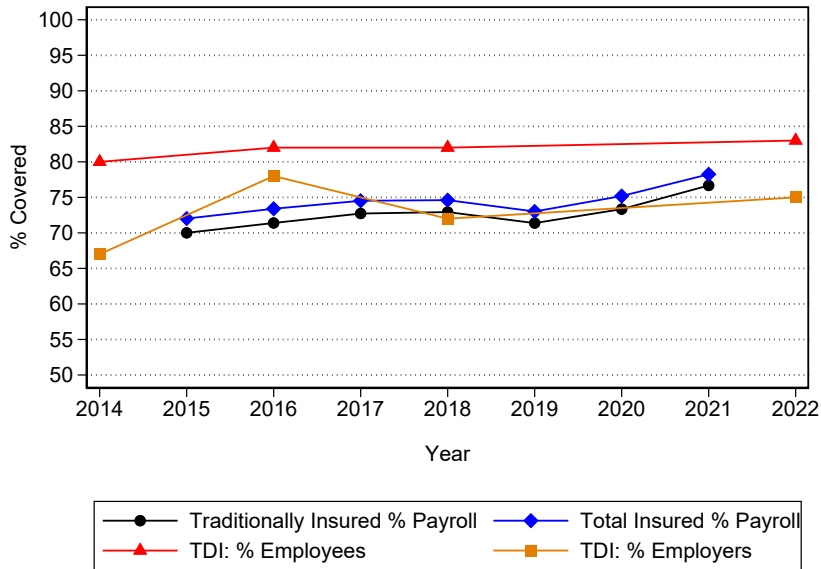
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Appendix Tables and Figures

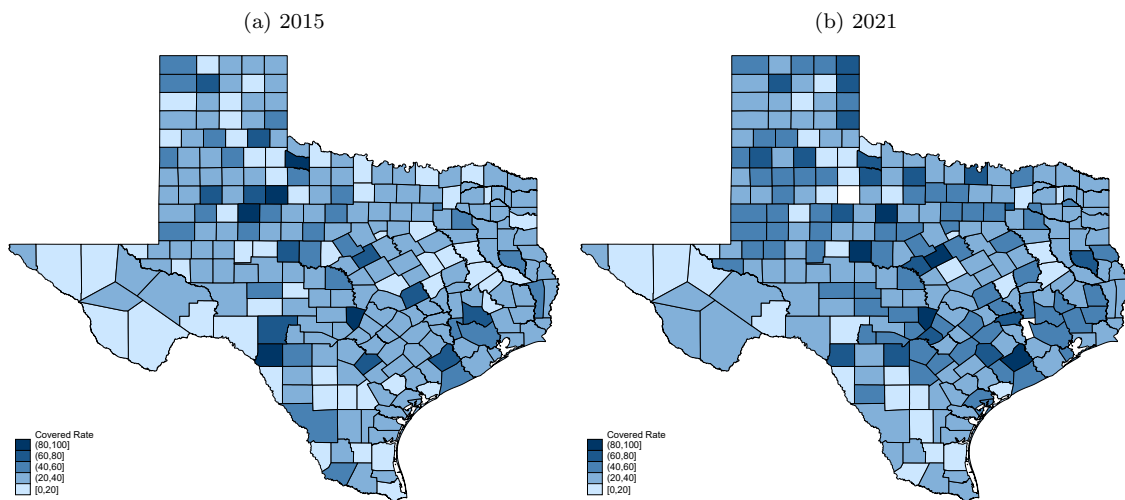
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Figure A.1: Covered Rates Over Time



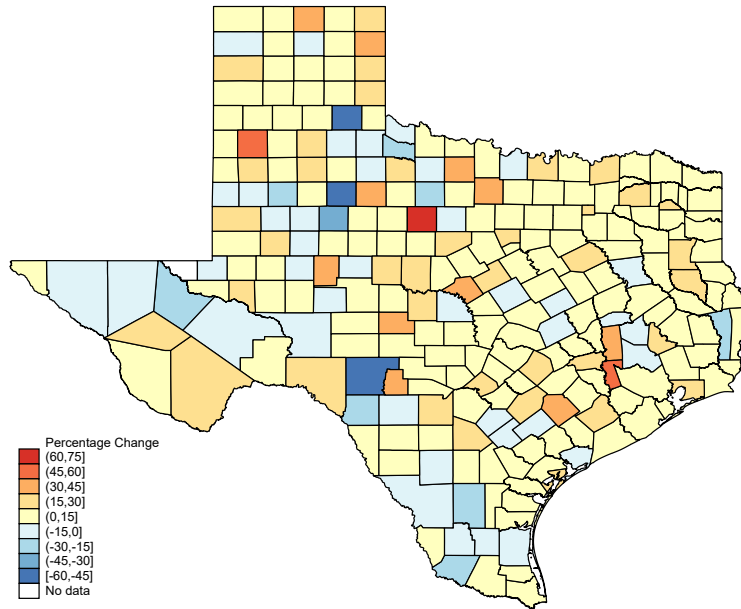
Notes: This figure displays covered rates over time, with and without self-insured employers included. It also includes publicly-reported information from a TDI subscriber survey to verify. Sources: Texas open records requests and QCEW, 2015-2021; [DWC \(2022\)](#).

Figure A.2: Covered Rates by County



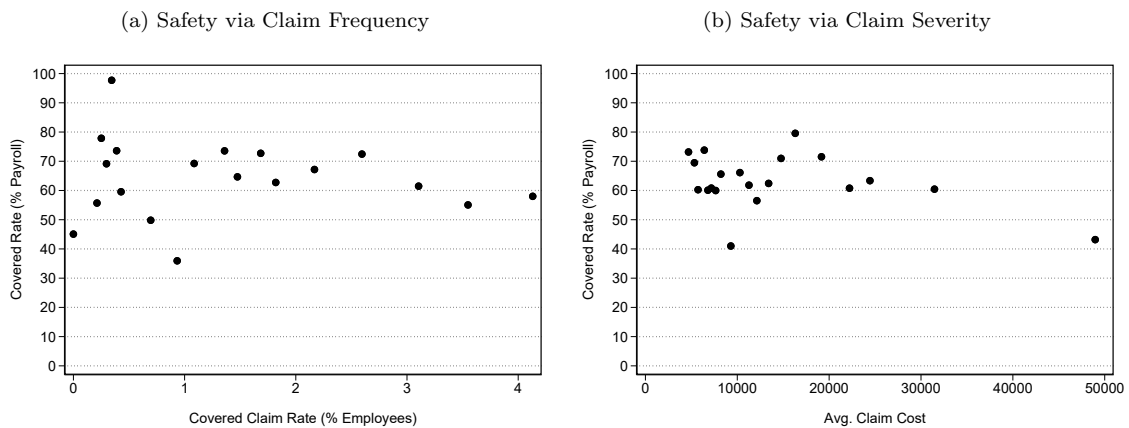
Notes: This figure displays covered rates by county in 2015 and 2021. The darker the blue, the higher the covered rate. Sources: Texas open records requests and QCEW, 2015-2021.

Figure A.3: Covered Rates by County, Temporal Changes



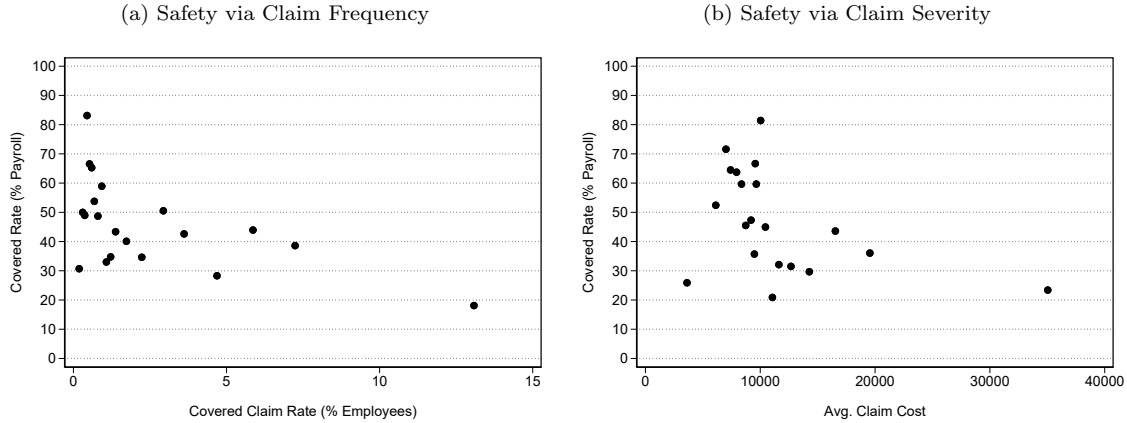
Notes: This figure displays percentage changes in covered rates between 2015 and 2021 by county. Sources: Texas open records requests and QCEW, 2015-2021.

Figure A.4: Covered Rates and Industry Safety (two-digit)



Notes: These figures display binned scatterplots of covered rates within 20 equally-sized bins of industries based on their covered claim rates in panel (a) and covered claim costs in panel (b). Industry is measured at the two-digit NAICS level. Sources: Texas open records requests and QCEW, 2015-2021.

Figure A.5: Covered Rates and Industry Safety (four-digit)



Notes: These figures display binned scatterplots of covered rates within 20 equally-sized bins of industries based on their covered claim rates in panel (a) and covered claim costs in panel (b). Industry is measured at the four-digit NAICS level. Sources: Texas open records requests and QCEW, 2015-2021.

Table A.1: Demand at the Insurer Level

	(1)	(2)	(3)	(4)	(5)
	ln(Covered Payroll)	ln(Covered Payroll)	ln(Covered Payroll)	ln(Number of Policies)	ln(Number of Occupations)
ln(Loadings)	-0.452*** (0.088)	-0.948*** (0.099)	-0.939*** (0.093)	-0.252 (0.185)	-0.259** (0.117)
ln(Loss Costs)			-0.691*** (0.019)		
Insurer Fixed Effects	Yes	Yes	Yes	Yes	Yes
Policy-Year Fixed Effects	Yes	No	Yes	Yes	Yes
Occupation by Policy-Year Fixed Effects	No	Yes	No	No	No
Observations	199,845	199,786	199,845	2,686	2,686

Notes: This table provides results for estimating regressions of Equation 5's design. Standard errors are clustered at the insurer level. Source: Texas open records requests, 2015-2021. Statistical significance is given by: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.